

Stochastic Calculus and Option Pricing

Synthesis Exercise Corrections

1 Week 1 — Option Pricing and Core Probability Tools

Part I — Random Variables and Distributions

Exercise 1.1 — A financial payoff as a random variable

- (a) The underlying is the stock, the maturity is $T = 1$, and the strike is $K = 100$. The possible payoff values are 0, 5, and 40.
- (b) The law of X is

$$\mathbb{P}(X = 0) = \frac{1}{4}, \quad \mathbb{P}(X = 5) = \frac{1}{2}, \quad \mathbb{P}(X = 40) = \frac{1}{4}.$$

Hence

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}, & 0 \leq x < 5, \\ \frac{3}{4}, & 5 \leq x < 40, \\ 1, & x \geq 40. \end{cases}$$

- (c) Direct calculation gives

$$\mathbb{E}[X] = \frac{1}{2}(5) + \frac{1}{4}(40) = 12.5, \quad \mathbb{E}[X^2] = \frac{1}{2}(25) + \frac{1}{4}(1600) = 412.5.$$

Therefore

$$\text{Var}(X) = 412.5 - 12.5^2 = 256.25, \quad \text{SD}(X) = \sqrt{256.25} \approx 16.01.$$

- (d) The discounted physical expectation is

$$e^{-0.03}\mathbb{E}[X] = 12.5e^{-0.03} \approx 12.13.$$

It uses physical probabilities. A no-arbitrage price is instead fixed by a replicating strategy, or equivalently by risk-neutral probabilities when the claim is replicable.

Exercise 1.2 — Cdf, density, and moments

- (a)

$$F_U(u) = \begin{cases} 0, & u < -1, \\ \frac{u+1}{4}, & -1 \leq u < 3, \\ 1, & u \geq 3, \end{cases} \quad f_U(u) = \frac{1}{4}\mathbf{1}_{[-1,3]}(u).$$

(b) From the cdf, $\mathbb{P}(0 < U \leq 2) = F_U(2) - F_U(0) = 3/4 - 1/4 = 1/2$. From the density, $\int_0^2 (1/4) du = 1/2$.

(c)

$$\mathbb{E}[U] = \frac{1}{4} \int_{-1}^3 u \, du = 1, \quad \mathbb{E}[U^2] = \frac{1}{4} \int_{-1}^3 u^2 \, du = \frac{7}{3},$$

so $\text{Var}(U) = 7/3 - 1 = 4/3$.

(d) Since $V = 2U + 1$, $V \sim \text{Unif}[-1, 7]$ and

$$F_V(v) = F_U\left(\frac{v-1}{2}\right) = \begin{cases} 0, & v < -1, \\ \frac{v+1}{8}, & -1 \leq v < 7, \\ 1, & v \geq 7. \end{cases}$$

Also $\mathbb{E}[V] = 2\mathbb{E}[U] + 1 = 3$ and $\text{Var}(V) = 4\text{Var}(U) = 16/3$.

Exercise 1.3 — Empirical moments, LLN, and CLT

(a) The sample mean is

$$\bar{x} = \frac{2 + 5 + 4 + 3 + 6}{5} = 4.$$

The sum of squared deviations is $4 + 1 + 0 + 1 + 4 = 10$. Hence the empirical variance is $v_5 = 10/5 = 2$, while the unbiased sample variance is $s_5^2 = 10/(5-1) = 2.5$.

(b) Since $\mathbb{E}[X_1] = 4 < \infty$, the LLN gives $\bar{X}_n \rightarrow 4$ in probability. The average daily count stabilizes near the population mean as the number of days grows.

(c) For a Poisson(4) variable, $\text{Var}(X_1) = 4$. Thus

$$\frac{\sqrt{n}(\bar{X}_n - 4)}{2} \Rightarrow \mathcal{N}(0, 1).$$

For $n = 100$, the approximate mean is 4 and the standard deviation is $2/\sqrt{100} = 0.2$.

(d)

$$\mathbb{P}(\bar{X}_{100} > 4.3) \approx \mathbb{P}\left(Z > \frac{4.3 - 4}{0.2}\right) = 1 - \Phi(1.5) \approx 0.0668.$$

(e) The LLN identifies the limit of the average, whereas the CLT gives the approximate distribution and $1/\sqrt{n}$ scale of its remaining error.

Exercise 1.4 — Characteristic function of a Gaussian

(a) Since the standard Gaussian density is $(2\pi)^{-1/2}e^{-z^2/2}$,

$$\varphi_Z(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iuz} e^{-z^2/2} \, dz.$$

At $u = 0$, the integral is the total mass of the density, so $\varphi_Z(0) = 1$.

(b) Differentiation under the integral sign gives

$$\varphi'_Z(u) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} iz e^{iuz} e^{-z^2/2} dz.$$

This interchange is justified, for example, by dominated convergence because $|z|e^{-z^2/2}$ is integrable.

(c) Since $ze^{-z^2/2} = -\frac{d}{dz}e^{-z^2/2}$, integration by parts yields

$$\begin{aligned} \varphi'_Z(u) &= -\frac{i}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{iuz} \frac{d}{dz} (e^{-z^2/2}) dz \\ &= -\frac{i}{\sqrt{2\pi}} \left(\left[e^{iuz} e^{-z^2/2} \right]_{-\infty}^{\infty} - iu \int_{-\infty}^{\infty} e^{iuz} e^{-z^2/2} dz \right). \end{aligned}$$

Because $|e^{iuz}| = 1$ and $e^{-z^2/2} \rightarrow 0$ as $|z| \rightarrow \infty$, the boundary term is zero. Therefore

$$\varphi'_Z(u) = -u\varphi_Z(u).$$

(d) The differential equation can be written $(e^{u^2/2}\varphi_Z(u))' = 0$. Hence

$$\varphi_Z(u) = Ce^{-u^2/2}.$$

The condition $\varphi_Z(0) = 1$ gives $C = 1$, and thus

$$\boxed{\varphi_Z(u) = e^{-u^2/2}}.$$

(e) Using $X = \mu + \sigma Z$,

$$\begin{aligned} \varphi_X(u) &= \mathbb{E}[e^{iu(\mu+\sigma Z)}] = e^{i\mu u} \mathbb{E}[e^{i(\sigma u)Z}] \\ &= e^{i\mu u} \varphi_Z(\sigma u) = \boxed{\exp\left(i\mu u - \frac{1}{2}\sigma^2 u^2\right)}. \end{aligned}$$

(f) If $X_j \sim \mathcal{N}(\mu_j, \sigma_j^2)$ for $j = 1, 2$ and the variables are independent, then

$$\begin{aligned} \varphi_{X_1+X_2}(u) &= \varphi_{X_1}(u)\varphi_{X_2}(u) \\ &= \exp\left(i(\mu_1 + \mu_2)u - \frac{1}{2}(\sigma_1^2 + \sigma_2^2)u^2\right). \end{aligned}$$

This is the characteristic function of

$$X_1 + X_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

Part II — Joint Distributions and Information

Exercise 1.5 — Reading a joint law

(a) The marginals are

$$\mathbb{P}(X = 0) = \mathbb{P}(X = 1) = 0.50, \quad \mathbb{P}(Y = L) = 0.30, \quad \mathbb{P}(Y = H) = 0.70.$$

(b) They are not independent because, for example, $\mathbb{P}(X = 1, Y = L) = 0.20 \neq 0.50 \times 0.30 = 0.15$.

(c)

$$\mathbb{P}(X = 1 \mid Y = L) = \frac{0.20}{0.30} = \frac{2}{3}, \quad \mathbb{P}(X = 1 \mid Y = H) = \frac{0.30}{0.70} = \frac{3}{7}.$$

(d) The signal L carries the larger conditional loss probability.

Exercise 1.6 — Information generated by a signal

Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$.

- (a) The level-set partition is $\{A, B\}$ and $\mathcal{G} = \{\emptyset, A, B, \Omega\}$.
- (b) Z_1 , Z_2 , and Z_4 are constant on both A and B , so they are $\mathcal{G}$ -measurable. The variable Z_3 is not: on B it equals 0 at outcome 4 and 1 at outcomes 5, 6.
- (c) One may take $g(y) = (-1)^y = 1 - 2y$ on $\{0, 1\}$.
- (d) Observing $Y = 1$ identifies only the block B ; it does not distinguish outcome 4 from outcomes 5 and 6.

Part III — Conditional Expectation

Exercise 1.7 — Conditional expectation on a finite partition

- (a) X is not $\mathcal{G}$ -measurable because it takes values 0 and 6 on the same atom $M = \{3, 4\}$.
- (b) The values of X on outcomes 1, ..., 6 are 0, 0, 0, 6, 6, 6, so $\mathbb{E}[X] = 3$. The atom averages are 0 on L , 3 on M , and 6 on H .
- (c)

$$\mathbb{E}[X \mid \mathcal{G}] = 0\mathbf{1}_L + 3\mathbf{1}_M + 6\mathbf{1}_H.$$

- (d) On L , both integrals are 0. On M , $\mathbb{E}[X\mathbf{1}_M] = 6/6 = 1 = 3\mathbb{P}(M)$. On H , $\mathbb{E}[X\mathbf{1}_H] = 12/6 = 2 = 6\mathbb{P}(H)$. Thus the defining averaging property holds on every atom and therefore on every event in $\mathcal{G}$.
- (e)

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}]] = 0 \cdot \frac{1}{3} + 3 \cdot \frac{1}{3} + 6 \cdot \frac{1}{3} = 3 = \mathbb{E}[X].$$

Exercise 1.8 — Conditional density and $\mathbb{E}[f(X) \mid Y]$

- (a)

$$\int_0^1 \int_0^1 (x + y) \, dx \, dy = 1.$$

The marginals are

$$f_X(x) = \left(x + \frac{1}{2}\right) \mathbf{1}_{(0,1)}(x), \quad f_Y(y) = \left(y + \frac{1}{2}\right) \mathbf{1}_{(0,1)}(y).$$

- (b) For $0 < y < 1$,

$$f_{X|Y}(x \mid y) = \frac{x + y}{y + 1/2} \mathbf{1}_{(0,1)}(x).$$

This conditional density depends on y , so X and Y are not independent.

- (c)

$$\mathbb{E}[X \mid Y = y] = \frac{\int_0^1 x(x + y) \, dx}{y + 1/2} = \frac{1/3 + y/2}{y + 1/2}.$$

Hence

$$\mathbb{E}[X \mid Y] = \frac{1/3 + Y/2}{Y + 1/2}.$$

(d) Similarly,

$$\mathbb{E}[X^2 \mid Y = y] = \frac{\int_0^1 x^2(x+y) dx}{y+1/2} = \frac{1/4 + y/3}{y+1/2}.$$

Thus $g(y) = (1/4 + y/3)/(y+1/2)$ and $\mathbb{E}[X^2 \mid Y] = g(Y)$. It is $\sigma(Y)$ -measurable because it is a measurable function of Y .

(e) Directly,

$$\mathbb{E}[X] = \int_0^1 x \left(x + \frac{1}{2}\right) dx = \frac{1}{3} + \frac{1}{4} = \frac{7}{12}.$$

Using the marginal density of Y ,

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \int_0^1 \frac{1/3 + y/2}{y+1/2} \left(y + \frac{1}{2}\right) dy = \int_0^1 \left(\frac{1}{3} + \frac{y}{2}\right) dy = \frac{7}{12}.$$

Exercise 1.9 — Gaussian conditioning

(a) With $a = \text{Cov}(X, Y) / \text{Var}(Y)$,

$$\text{Cov}(R, Y) = \text{Cov}(X, Y) - a \text{Var}(Y) = 0.$$

(b) (R, Y) is jointly Gaussian because it is an affine transformation of (X, Y) . Jointly Gaussian and uncorrelated variables are independent.

(c) Since $X = \mu_X + a(Y - \mu_Y) + R$ and $\mathbb{E}[R] = 0$,

$$\mathbb{E}[X \mid Y] = \mu_X + \frac{\text{Cov}(X, Y)}{\text{Var}(Y)}(Y - \mu_Y) = \mu_X + \rho \frac{\sigma_X}{\sigma_Y}(Y - \mu_Y).$$

Also

$$\text{Var}(X \mid Y) = \text{Var}(R) = \sigma_X^2 - \frac{\text{Cov}(X, Y)^2}{\sigma_Y^2} = \sigma_X^2(1 - \rho^2).$$

(d) A larger $|\rho|$ makes Y more informative about X and reduces the residual variance. When $|\rho|$ is close to one, the signal is an effective hedge; when it is close to zero, little risk is explained.

Part IV — Stochastic Processes and Filtrations

Exercise 1.10 — Three coin tosses viewed through time

(a) $\mathcal{F}_0$ has the single atom Ω . At the next dates,

$$\begin{aligned} \text{Atoms}(\mathcal{F}_1) &= \{H^{**}, T^{**}\}, \\ \text{Atoms}(\mathcal{F}_2) &= \{HH^*, HT^*, TH^*, TT^*\}, \end{aligned}$$

while $\mathcal{F}_3$ has the eight singleton outcomes. Each new observation refines the preceding partition, so

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \mathcal{F}_3.$$

(b) S_n is determined by the first n tosses, hence is $\mathcal{F}_n$ -measurable for every n .

- (c) A_2 depends on the third toss and is not $\mathcal{F}_2$ -measurable. The other values are measurable at their respective times. The failure at time 2 is look-ahead.
- (d) The capped time is 1 on $H **$, 2 on $TH*$, and 3 on $TT*$. Moreover

$$\{\tau \leq 0\} = \emptyset, \quad \{\tau \leq 1\} = H **, \quad \{\tau \leq 2\} = \Omega \setminus TT*, \quad \{\tau \leq 3\} = \Omega,$$

and each event belongs to the corresponding sigma-algebra.

Part V — Martingales and Fair Games

Exercise 1.11 — A fair but asymmetric random walk

(a)

$$\mathbb{E}[\xi_k] = 2 \cdot \frac{1}{3} - 1 \cdot \frac{2}{3} = 0, \quad \mathbb{E}[\xi_k^2] = 4 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = 2.$$

The expected gain on each new play is zero, even though the possible gains are asymmetric.

(b) Since S_n is $\mathcal{F}_n$ -measurable and ξ_{n+1} is centered and independent of $\mathcal{F}_n$,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = S_n + \mathbb{E}[\xi_{n+1} \mid \mathcal{F}_n] = S_n.$$

(c) Expanding the square gives

$$\begin{aligned} \mathbb{E}[S_{n+1}^2 - 2(n+1) \mid \mathcal{F}_n] &= S_n^2 + 2S_n\mathbb{E}[\xi_{n+1}] + \mathbb{E}[\xi_{n+1}^2] - 2n - 2 \\ &= S_n^2 - 2n. \end{aligned}$$

(d) With up probability $1/4$, $\mathbb{E}[\xi_k] = 2/4 - 3/4 = -1/4$. Thus S_n has predictable drift $-n/4$ and

$$M_n = S_n + \frac{n}{4}$$

is a martingale.

(e) Conditional on the entire observed history, expected wealth after the next fair play equals current wealth. Past wins or losses do not create a predictable future gain.

Exercise 1.12 — Replication and risk-neutral pricing

(a)

	up	down
S_1	65	45
$H = (S_1 - 55)^+$	10	0

(b) The stock-pricing equation is

$$50 = \frac{1}{1.05}(65q + 45(1 - q)),$$

so $52.5 = 45 + 20q$ and $q = 3/8 \in (0, 1)$.

(c)

$$V_0 = \frac{1}{1.05} \left(10 \cdot \frac{3}{8} \right) = \frac{25}{7} \approx 3.57.$$

(d) Solving $65\Delta + B_1 = 10$ and $45\Delta + B_1 = 0$ gives

$$\Delta = \frac{1}{2}, \quad B_1 = -22.5.$$

(e) The initial replication cost is

$$\Delta S_0 + \frac{B_1}{R} = 25 - \frac{22.5}{1.05} = \frac{25}{7} = V_0.$$

The physical up probability does not enter because the terminal payoff is matched state by state.

(f) Under the risk-neutral probability, the discounted stock price satisfies $S_0 = \mathbb{E}^q[S_1/R]$. The same martingale-pricing restriction gives $V_0 = \mathbb{E}^q[H/R]$ for the replicable claim.

Part VI — Mathematical Toolbox

Exercise 1.13 — Derivatives, gradients, and the chain rule

(a)

$$\frac{(x+h)^2 - x^2}{h} = 2x + h \longrightarrow 2x,$$

so $g'(x) = 2x$.

(b) For $f(x, y) = x^2y + e^y$,

$$f_x = 2xy, \quad f_y = x^2 + e^y, \quad \nabla f = (2xy, x^2 + e^y).$$

The second derivatives are $f_{xx} = 2y$, $f_{xy} = f_{yx} = 2x$, and $f_{yy} = e^y$, hence

$$\nabla^2 f(x, y) = \begin{pmatrix} 2y & 2x \\ 2x & e^y \end{pmatrix}.$$

(c) The multivariable chain rule gives

$$h'(t) = V_t(t, S(t)) + V_s(t, S(t))S'(t).$$

(d) For $V(t, s) = e^{-rt}s^2$,

$$V_t = -re^{-rt}s^2, \quad V_s = 2e^{-rt}s,$$

and therefore

$$h'(t) = e^{-rt}(-rS(t)^2 + 2S(t)S'(t)).$$

(e) The payoff is not differentiable at $s = K$: the left derivative is 0 and the right derivative is 1. Away from K ,

$$v'(s) = \begin{cases} 0, & s < K, \\ 1, & s > K. \end{cases}$$

2 Week 2 — Brownian Motion, Stopping Times, and Quadratic Variation

Part I — Gaussian Structure and Brownian Transformations

Exercise 2.1 — Gaussian vectors and the covariance kernel

- (a) Independent increments give

$$(W_r, W_s - W_r, W_t - W_s) \sim \mathcal{N}(0, \text{diag}(r, s - r, t - s)).$$

- (b) The level vector is a linear transformation of this Gaussian vector, hence is Gaussian, with covariance matrix

$$\begin{pmatrix} r & r & r \\ r & s & s \\ r & s & t \end{pmatrix}.$$

- (c) Since $W_t = W_s + (W_t - W_s)$,

$$W_t \mid \mathcal{F}_s \sim \mathcal{N}(W_s, t - s), \quad W_t \mid (W_s = x) \sim \mathcal{N}(x, t - s).$$

- (d) For any dates and coefficients,

$$\sum_{i,j} a_i a_j K(t_i, t_j) = \text{Var}\left(\sum_i a_i W_{t_i}\right) \geq 0.$$

- (e) Every finite vector of Brownian values is centered Gaussian. A Gaussian law is determined by its mean and covariance matrix, whose entries are generated by K .

Exercise 2.2 — Symmetry, scaling, renewal, and time reversal

- (a) Both processes start from zero and are continuous. Their increments over disjoint intervals are independent, and

$$X_t - X_u \sim \mathcal{N}(0, t - u), \quad Y_t - Y_u = \frac{W_{ct} - W_{cu}}{\sqrt{c}} \sim \mathcal{N}(0, t - u).$$

Hence both are standard Brownian motions.

- (b) $Z_0 = 0$, and its increments are Brownian increments after time s . They are independent of one another and of $\mathcal{F}_s$, Gaussian with the correct variances, and continuous.

- (c) For $0 \leq u < t \leq s$,

$$R_t - R_u = W_{s-t} - W_{s-u} \sim \mathcal{N}(0, t - u).$$

Disjoint reversed intervals yield independent increments. Together with continuity, this proves equality of the finite-dimensional laws.

- (d) Applying scaling to the whole path,

$$(W_{cu})_{0 \leq u \leq t} \stackrel{d}{=} (\sqrt{c}W_u)_{0 \leq u \leq t}.$$

Taking maxima gives the required identity.

(e) Almost surely,

$$\begin{aligned} \limsup_{t \rightarrow \infty} \frac{W_t}{\sqrt{2t \log \log t}} &= 1, & \liminf_{t \rightarrow \infty} \frac{W_t}{\sqrt{2t \log \log t}} &= -1, \\ \limsup_{t \downarrow 0} \frac{W_t}{\sqrt{2t \log \log(1/t)}} &= 1, & \liminf_{t \downarrow 0} \frac{W_t}{\sqrt{2t \log \log(1/t)}} &= -1. \end{aligned}$$

Part II — Markov Property, Martingales, and Stopping

Exercise 2.3 — Markov semigroup and martingale correction

(a) Write $W_{t+h} = W_t + G_h$, where $G_h \sim \mathcal{N}(0, h)$ is independent of $\mathcal{F}_t$. Conditioning gives

$$\mathbb{E}[f(W_{t+h}) \mid \mathcal{F}_t] = (P_h f)(W_t).$$

(b) Applying the Markov identity twice,

$$\mathbb{E}[f(W_{t+h+k}) \mid \mathcal{F}_t] = P_h(P_k f)(W_t) = P_{h+k}f(W_t),$$

so $P_{h+k} = P_h P_k$.

(c) Independent centered increments yield

$$\mathbb{E}[W_t \mid \mathcal{F}_s] = W_s, \quad \mathbb{E}[W_t^2 \mid \mathcal{F}_s] = W_s^2 + (t - s).$$

(d) The first identity proves that W is a martingale. The second gives

$$\mathbb{E}[W_t^2 - t \mid \mathcal{F}_s] = W_s^2 - s,$$

so $W_t^2 - t$ is also a martingale.

(e) For every continuous square-integrable martingale X , there is a unique increasing predictable process $\langle X \rangle$, starting from zero, such that $X_t^2 - \langle X \rangle_t$ is a martingale. For Brownian motion, $\langle W \rangle_t = [W]_t = t$.

Exercise 2.4 — Stopping times and information at stopping

(a) The deterministic time T , the first hit τ_a , and the first exit $\tau_{\ell, u}$ are stopping times. Continuity makes their occurrence detectable from the observed path. The last zero ρ_T and the time of the maximum γ_T require future information and are not stopping times.

(b) By continuity, the path hits a by time t exactly when its maximum over $[0, t]$ is at least a .

(c)

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t \text{ for every } t \geq 0\}.$$

For continuous adapted W , the stopped variable W_τ is obtained as a limit of values at simple stopping times, so it is $\mathcal{F}_\tau$ -measurable.

(d) Strong Markov says that

$$(W_{\tau+v} - W_\tau)_{v \geq 0}$$

is Brownian motion independent of $\mathcal{F}_\tau$. At τ_a , the fresh process is $(W_{\tau_a+v} - a)_{v \geq 0}$.

(e) The reflection starts at the random time τ_a . Deterministic Markov does not justify renewal there; strong Markov does, and symmetry then permits the sign flip of the fresh continuation.

Part III — Reflection and Hitting Times

Exercise 2.5 — Reflection principle and first passage

(a) On $\{\tau_a \leq t\}$, define

$$W_u^* = \begin{cases} W_u, & u \leq \tau_a, \\ 2a - W_u, & u > \tau_a. \end{cases}$$

This pairs $\{M_t \geq a, W_t < a\}$ with $\{W_t > a\}$.

(b) Hence, for $a > 0$,

$$\mathbb{P}(M_t \geq a) = \mathbb{P}(\tau_a \leq t) = 2 \left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right) \right).$$

Thus $\mathbb{P}(M_t \leq a) = 2\Phi(a/\sqrt{t}) - 1$.

(c)

$$f_{\tau_a}(t) = \frac{a}{\sqrt{2\pi t^3}} \exp\left(-\frac{a^2}{2t}\right), \quad t > 0.$$

(d) The tail of M_t equals that of $|W_t|$, so

$$M_t \stackrel{d}{=} |W_t|, \quad \mathbb{E}[M_t] = \mathbb{E}[|W_t|] = \sqrt{\frac{2t}{\pi}}.$$

(e) By scaling,

$$\mathbb{P}(\tau_a \leq t) = 2 \left(1 - \Phi\left(\frac{a}{\sqrt{t}}\right) \right) = \mathbb{P}(a^2 \tau_1 \leq t).$$

(f) Letting $t \rightarrow \infty$ in the cdf gives $\mathbb{P}(\tau_a < \infty) = 1$. Moreover,

$$\mathbb{P}(\tau_a > t) = 2\Phi(a/\sqrt{t}) - 1 \sim a\sqrt{\frac{2}{\pi t}},$$

whose integral over large t diverges. Hence $\mathbb{E}[\tau_a] = \infty$.

Part IV — Quadratic Variation

Exercise 2.6 — Realized quadratic variation

(a) The increments are independent $\mathcal{N}(0, T/n)$, so

$$\mathbb{E}[Q_n] = T, \quad \text{Var}(Q_n) = n 2(T/n)^2 = \frac{2T^2}{n}.$$

(b) Therefore $\mathbb{E}[(Q_n - T)^2] = 2T^2/n \rightarrow 0$.

(c) By the mean value theorem,

$$\sum_i (\Delta_i A)^2 \leq \|A'\|_\infty^2 T |\pi_n| \rightarrow 0.$$

(d) Cauchy–Schwarz gives

$$\left| \sum_i \Delta_i A \Delta_i W \right| \leq \left(\sum_i (\Delta_i A)^2 \right)^{1/2} Q_n^{1/2}.$$

The first factor tends to zero and $Q_n \rightarrow T$, so the product tends to zero in probability.

(e) Expanding squared increments and using the preceding limits yields $[A + \sigma W]_T = \sigma^2 T$.

(f) A consistent estimator is

$$\hat{\sigma}_n^2 = \frac{1}{T} \sum_{i=0}^{n-1} (X_{t_{i+1}} - X_{t_i})^2 \xrightarrow{\mathbb{P}} \sigma^2.$$

Part V — Price Models and Trading Gains

Exercise 2.7 — Symbolic analysis of geometric Brownian motion

(a)

$$\log(S_t/S_0) \sim \mathcal{N}\left((\mu - \tfrac{1}{2}\sigma^2)t, \sigma^2 t\right),$$

and, conditionally on $\mathcal{F}_s$,

$$\log(S_t/S_s) \sim \mathcal{N}\left((\mu - \tfrac{1}{2}\sigma^2)(t-s), \sigma^2(t-s)\right).$$

(b) Lognormal moments give

$$\mathbb{E}[S_t] = S_0 e^{\mu t}, \quad \text{Var}(S_t) = S_0^2 e^{2\mu t} (e^{\sigma^2 t} - 1),$$

and

$$\text{Median}(S_t) = S_0 e^{(\mu - \sigma^2/2)t}.$$

(c)

$$\mathbb{P}(S_t > K) = 1 - \Phi\left(\frac{\log(K/S_0) - (\mu - \sigma^2/2)t}{\sigma\sqrt{t}}\right).$$

(d) Since $\log S_t$ is a smooth drift plus σW_t , $[\log S]_t = \sigma^2 t$. The drift has zero quadratic variation.

(e) If $R_i = \log(S_{t_{i+1}}/S_{t_i})$, then

$$\frac{1}{T} \sum_i R_i^2 \xrightarrow{\mathbb{P}} \sigma^2$$

as the mesh tends to zero.

Exercise 2.8 — A two-step adapted trading gain

(a) At time t_1 , Δ_1 may use the path already observed but not the future increment $W_T - W_{t_1}$.

(b)

$$\mathbb{E}[\Delta_1(W_T - W_{t_1})] = \mathbb{E}[\Delta_1 \mathbb{E}[W_T - W_{t_1} \mid \mathcal{F}_{t_1}]] = 0.$$

Also $\mathbb{E}[W_{t_1}] = 0$, hence $\mathbb{E}[G_T] = 0$.

(c) The cross-term is zero because

$$\mathbb{E}[W_{t_1} \Delta_1 (W_T - W_{t_1})] = 0.$$

Moreover,

$$\mathbb{E}[\Delta_1^2 (W_T - W_{t_1})^2] = (T - t_1) \mathbb{E}[\Delta_1^2].$$

Therefore

$$\mathbb{E}[G_T^2] = \Delta_0^2 t_1 + (T - t_1) \mathbb{E}[\Delta_1^2].$$

(d) The proposed position depends on the sign of a future increment and is not $\mathcal{F}_{t_1}$ -measurable.

(e) Set $\Delta_t = \Delta_0$ on $[0, t_1]$ and $\Delta_t = \Delta_1$ on $(t_1, T]$. Then

$$G_T = \int_0^T \Delta_t \, dW_t.$$